

Jo-Yao Hsu (JY Hsu)

(+886) 916631438 | jyhsu506@gmail.com | <https://joyao.github.io>

PROFESSIONAL EXPERIENCE

Google | Test Engineer, Pixel SoC Software Security

September 2025 - Present

- **Firmware Security Validation:** Engineered high-throughput, host-driven Python test automation frameworks targeting low-level Pixel SoC SecureBoot flows and cryptographic verification, integrating directly into CI/CQ pre-submit pipelines to guarantee root-of-trust integrity and prevent regression vulnerabilities.
- **Security & Manufacturing Provisioning Infrastructure:** Designed automated CI pipelines for secure provisioning, hardware anti-rollback enforcement, and key/binary synchronization across factory and post-factory stages.
- **Scaled Distributed Hardware Lab Infra:** Utilized gRPC services to integrate hardware lab APIs with device management platforms, automating UART log capturing and remote device control to eliminate environmental test flakiness.
- **Advanced Developer Test Tooling:** Extended Mobly-based host test frameworks with custom CLI tools, smart-flashing capabilities, and automated diagnostics for local device testing, boosting engineer onboarding and debugging efficiency.
- **AI-Driven Failure Triaging:** Pioneered AI-assisted log analysis and automated failure triaging pipelines to accelerate root-cause analysis, successfully reducing test flake rates toward zero across SoC areas.
- **AI-Driven Test & Code Coverage (2026 A²PS Hackathon Top 10 Finalist in Track):** Architected and built a quick PoC system to analyze and map firmware test coverage; proposed intelligent test case selection for future runs.

DELL Technologies | Senior Software Firmware Engineer

August 2021 - August 2025

Automated CI/CQ Workflow Implemented for Firmware Developing, Testing and Releasing as an SDET

- Led 20-member cross-functional teams across 8 teams, 4 companies, and 3 countries to develop firmware testing automation systems, covering 200+ test cases, a pivotal Dell implementation project in recent years.
- Implemented automated daily and quarterly BIOS builds and testing across 10+ platforms, enabling “Shift-Left Testing”, improving stability, and significantly reducing pre-release costs and post-release BIOS errors.
- Designed, developed, and implemented a scalable CI/CQ system for feature and integration automated testing, including BIOS cloud build, firmware signing, ODM build, and auto-release features, enhancing efficiency and reliability.
- Developed intuitive interfaces for automation systems, improving accessibility and usability for 8+ teams and stakeholders.
- **DELL FY23 Game Changer Award** for driving massive progress with a transformative BIOS automation workflow, fostering team and ODM ownership, and accelerating firmware test case integration, significantly enhancing development efficiency.
- **2025 DELL ISG Hackathon Special Award for AI-For:** Proposed an innovative solution using RAG with chatbot and LaaS to automate hardware leasing, firmware setup, and testing, cutting setup time by 80% and streamlining resource allocation.

BIOS/UEFI Feature Development & Platform Issue Resolution

- Co-own the design and development of the D4D (Design for Diagnosis) feature, increasing BIOS debugging log capacity by 15x, implementing standardized logging, critical debugging table storage, compression, and data sorting for future AI.
- Developed an OS-based tool for decompressing and decoding BIOS debug logs from customer TSR logs, streamlining error analysis and enhancing troubleshooting efficiency.
- Mentored interns in utilizing BIOS debug logs for organization and analysis, and spearheaded the establishment of an AI model for preliminary detection of BIOS issues.
- **DELL FY25 Game Changer Award** for groundbreaking innovation on the D4D feature, enabling full boot log capture with minimal POST time impact.

CECI Engineering Consultants, Inc. | Technical Leader and Software Engineer

July 2018 - August 2021

- **Full-Stack Leadership:** Directed a 5-person Agile development team to architect Surface Deformation and Urban Planning systems, successfully delivering 15+ large-scale web solutions for critical infrastructure projects.
- **CI/CD & Automated Testing Architecture:** Architected cross-network CI/CD pipelines and integrated automated third-party API testing, reducing release cycles by 90% and securing zero-regression product deliveries.
- **Database Optimization:** Overhauled daily Oracle database synchronization workflows and implemented address fuzzy matching, boosting spatial data fetching speed by 50% and spatial accuracy by 7%.

EDUCATION

National Taiwan University

Taipei, Taiwan

Master of Science in Civil Engineering (GIS and Surveying) Graduated with 1st Class Honors

September 2016 - June 2018

- Developed error-adjustment software with GUI & DLL, reducing costs by \$3K USD annually.

National Chengchi University

Taipei, Taiwan

Bachelor's Degree in Geomatics, Land Economics (Geomatics and Surveying) Graduated with 1st Class Honors

September 2012 - June 2016

ADDITIONAL SKILLS AND AWARDS

- **Core Technologies:** Python, JavaScript/TypeScript, C#, SQL, Node.js, React.js, C, .NET, UEFI/BIOS, Linux.
- **DevOps & Cloud:** Git, Jenkins, Docker, AWS, DevSecOps, SonarQube, Vault, CI/CD Architecture.
- **Certifications:** AWS Certified Solutions Architect, PMP Certification Training, Professional Engineer Certificate – Survey, Certified Procurement Professional.
- **Awards:** 1st Place Water OpenData IoT; 2024 & 2020 iTHelp Keep Sharing Award; Hacktoberfest 2024.
- **Open-Source & Languages:** Maintainer of hugo-toha theme; Contributor to github-readme-stats. Fluent in English and Native in Chinese.

